Avevamo concluso la scorsa volta con l’integrazione delle funzioni razionali fratte
Concludiamo questa parte con questi altri due esercizi :,

1) $\int \frac{x+1}{9x^2+4+12x}dx = \int \frac{x+1}{(3x+2)^2}dx$

$n = m - 1$ in questo caso per cui operiamo la sostituzione

$t = 3x+2 \qquad dt = 3dx \qquad dx = \frac{1}{3}dt$

$3x = t-2 \qquad x = \frac{1}{3}(t-2)$

e otteniamo:

$$\int \frac{1/3(t-2)+1}{t^2}\cdot\frac{1}{3}dt = \int \frac{\frac{1}{3}t-\frac{2}{3}+1}{3t^2}dt =$$

$$\int \frac{1}{9t}dt + \frac{1}{9}\int \frac{1}{t^2}dt = \frac{1}{9}\log|t| + \frac{1}{9}\left(-\frac{1}{t}\right)+c =$$

$$\frac{1}{9}\log|3x+2| - \frac{1}{9}\frac{1}{(3x+2)}+c.$$

2) $\int \frac{dx}{4x^2+4x+1} = \int \frac{dx}{(2x+1)^2} = \frac{1}{2}\int 2(2x+1)^{-2}dx =$

$$= \frac{1}{2}\frac{(2x+1)^{-1}}{-1}+c = -\frac{1}{2(2x+1)}+c.$$

Regola:

$$\int [f(x)]^{\alpha}\cdot f'(x) = \frac{[f(x)]^{\alpha+1}}{\alpha+1}+c$$

3) $\int \frac{1}{\sin 2x}dx =$

Regola $\sin^2 x + \cos^2 x = 1$

formula duplicazione del seno: $\sin 2x = 2\sin x\cos x$

$$= \int \frac{\sin^2 x + \cos^2 x}{2\sin x\cos}dx =$$

$$= \int \frac{\sin^{\cancel{2}} x}{2\cancel{\sin x}\cos x}dx + \int \frac{\cos^2 x}{2\sin x\cos x}dx =$$

$$= \frac{1}{2}\int \frac{\sin x}{\cos x}dx + \frac{1}{2}\int \frac{\cos x}{\sin x}dx = -\frac{1}{2}\log|\cos x| + \frac{1}{2}\log|\sin x| + c$$

Ricordando le regole dei logaritmi, si ha:

$$= \frac{1}{2}\log|\mathrm{tg}\, x| + c$$

4) $\int \frac{2x^2+3x-1}{x^3+2x^2+x+2}dx$

$n = m - 1$

Poichè il denominatore raccogliendo

$$(x+2) \to x^2(x+2)+(x+2) = (x^2+1)(x+2)$$

Otteniamo

$$\int \frac{A}{x+2}dx + \int \frac{Bx+C}{x^2+1}dx$$

$$\frac{A}{x+2} + \frac{Bx+C}{x^2+1} = \frac{2x^2+3x-1}{x^3+2x^2+x+2}$$

$$A(x^2+1) + (Bx+C)(x+2) = 2x^2+3x-1$$

$$Ax^2 + A + Bx^2 + 2Bx + Cx + 2C = 2x^2+3x-1$$

$$x^2(A+B) + x(2B+C) + A + 2C = 2x^2+3x-1$$

$$\begin{cases} A+B = 2 \\ 2B+C = 3 \\ A+2C = -1 \end{cases} \qquad \begin{cases} B-2C = 3 \\ 6+4C+C = 3 \\ A = 2-\frac{9}{5} = \frac{1}{5} \end{cases} \qquad \begin{matrix} B = 3+2C \\ B = 3-\frac{6}{5} = \frac{9}{5} \\ C = -\frac{3}{5} \end{matrix}$$

Quindi

$$\int \frac{2x^2+3x-1}{x^3+2x^2+x+2}dx = \frac{1}{5}\int \frac{1}{x+2}dx + \frac{3}{5}\int \frac{3x-1}{x^2+1}dx =$$

$$\frac{1}{5}\log|x+2| + \frac{9}{5}\frac{1}{2}\int \frac{2x-\frac{2}{3}}{x^2+1}dx =$$

$$\frac{1}{5}\log|x+2| + \frac{9}{5\cdot 2}\int \frac{2x}{x^2+1}dx - \frac{3}{5}\int \frac{dx}{x^2+1} =$$

$$\frac{1}{5}\log|x+2| + \frac{9}{10}\log|x^2+1| - \frac{3}{5}\mathrm{arctg}\, x + c$$

5) $\int \frac{x^2+1}{(x+1)^2(x^2-x+1)}dx$

Si procede nello stesso modo:

$$\frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{Cx+D}{x^2-x+1} = \frac{x^2+1}{(x+1)^2(x^2-x+1)}$$

Per esercizio per la prossima volta,
provate da soli lo risolviamo
poi in aula se ci fossero dubbi!

Metodo di integrazione per parti

$$\int \underbrace{f'(x)}_{\text{fattore differenziale}}\cdot\underbrace{g(x)}_{\text{fattore finito}}\,dx = f(x)\cdot g(x) - \int f(x)\cdot g'(x)dx$$

1) $\int x^3e^{-x^2}dx = -\frac{1}{2}\int (-2x)\,x^2\,e^{-x^2}dx = -\frac{1}{2}e^{-x^2}\cdot x^2 + \frac{1}{2}\int e^{-x^2}2x\,dx =$

$$= -\frac{1}{2}e^{-x^2}\cdot x^2 - \frac{1}{2}e^{-x^2} + c = -\frac{1}{2}e^{-x^2}(x^2+1) + c.$$

2) $-\int -x^3e^{-x}dx = -x^3e^{-x} - \int e^{-x}(-3x^2)dx =$

$$= -x^3e^{-x} - 3e^{-x}x^2 - 3\int -e^{-x}\cdot 2x\,dx =$$

$$= -x^3e^{-x} - 3x^2e^{-x} - 6e^{-x}\cdot x + 6\int e^{-x}dx =$$

$$= -x^3e^{-x} - 3x^2e^{-x} - 6e^{-x}x - 6e^{-x} + c =$$

$$= -e^{-x}(x^3+3x^2+6x+6) + c.$$

3) $\int e^x\sin x\,dx = e^x\sin x - \int e^x\cos x\,dx =$

$$= e^x\sin x - e^x\cos x + \int e^x(-\sin x)dx =$$

$$= e^x(\sin x - \cos x) - \int e^x\sin x\,dx$$

poichè ↗

Si ha:

$$2\int e^x\sin x\,dx = e^x(\sin x - \cos x) + c$$

$$\int e^x\sin x\,dx = \frac{e^x}{2}(\sin x - \cos x) + \tilde{c}$$

Ricordiamo che:

$$\int e^{\alpha x}\cos\beta x\,dx \qquad \int e^{\alpha x}\sin\beta x\,dx$$

$$\int e^{(\alpha+i\beta)x}dx = \frac{1}{\alpha+i\beta}\int(\alpha+i\beta)e^{(\alpha+i\beta)x}dx$$

$$= \frac{1}{\alpha+i\beta}e^{(\alpha+i\beta)x}$$

$$\int e^{\alpha x}\cdot e^{i\beta x}dx = \int e^{\alpha x}(\cos\beta x + i\sin\beta x)dx =$$

$$= \int e^{\alpha x}\cos\beta x\,dx + i\int e^{\alpha x}\sin\beta x\,dx =$$

$$= \frac{\alpha+i\beta}{\alpha^2+\beta^2}e^{\alpha x}\cdot e^{i\beta x} =$$

$$= \frac{\alpha-i\beta}{\alpha^2+\beta^2}e^{\alpha x}(\cos\beta x + i\sin\beta x) =$$

$$= \frac{\alpha}{\alpha^2+\beta^2}e^{\alpha x}\cos\beta x + i\frac{\alpha}{\alpha^2+\beta^2}e^{\alpha x}\sin\beta x - \frac{i\beta e^{\alpha x}}{\alpha^2+\beta^2}\cos\beta x$$

$$+ \frac{\beta e^{\alpha x}\sin\beta x}{\alpha^2+\beta^2} =$$

$$= \frac{e^{\alpha x}}{\alpha^2+\beta^2}(\alpha\cos\beta x + \beta\sin(\beta x)) + i\frac{e^{\alpha x}}{\alpha^2+\beta^2}(\alpha\sin\beta x - \beta\cos\beta x)$$

$$\int e^{\alpha x}\cos\beta x\,dx = \frac{e^{\alpha x}}{\alpha^2+\beta^2}(\alpha\cos\beta x + \beta\sin\beta x) + c$$

$$\int e^{\alpha x}\sin\beta x\,dx = \frac{e^{\alpha x}}{\alpha^2+\beta^2}(\alpha\sin\beta x - \beta\cos\beta x) + c$$

⇒ Per le funzioni goniometriche

$$\int f(\sin x)\cos x\,dx \qquad \int f(\cos x)\sin x\,dx$$

$u = \sin x \qquad du = +\cos x\,dx$ → segui!

$v = \cos x \qquad dv = -\sin x\,dx$

Es 1) $\int \sin^3 x\cos^2 x\,dx = \int \sin x\,\sin^2 x\cos^2 x\,dx =$

$\sin^{\alpha} x\cdot\cos^{\beta} x \qquad \alpha \neq \beta$

$$= \int \sin x(1-\cos^2 x)\cos^2 x\,dx =$$

$$= \int \sin x\cos^2 x\,dx - \int \sin x\cos^4 x\,dx$$

$v = \cos x \qquad dv = -\sin x\,dx$

$$= \int v^2(-dv) + \int v^4dv = -\frac{v^3}{3} + \frac{v^5}{5} + c$$

$$= \frac{1}{3}\cos^3 x + \frac{1}{5}\cos^5 x + c.$$